According to Neil's Bohr quantization rule, the angular momentum of an electron moving in a stationary orbit is quantized which means

$$L = n\hbar = n\frac{h}{2\pi}$$

where h is the Planck's constant (h = 6.63 x 10^{-34} J s) and number n is the orbit number, i.e., n = 1,2,3,..... Hence,

$$\mu_L = \frac{e}{2m}L = n\frac{eh}{4\pi m}$$

$$\mu_L = n \times \frac{(1.60\times10^{-19})h}{4\pi m}\,\mathrm{A\,m^2}$$

$$= n \times \frac{(1.60\times10^{-19})(6.63\times10^{-34})}{4\times3.14\times(9.11\times10^{-31})}$$

$$\mu_L = n\times9.27\times10^{-24}\,\mathrm{A\,m^2}$$

The minimum value of magnetic moment can be obtained by substituting n = 1,

$$\mu_L = 9.27\times10^{-24}\,\mathrm{A\,m^2} = 9.27\times10^{-24}\,\mathrm{J\,T^{-1}}$$
$$= (\mu_L)_{\min} = \mu_B$$

where $\mu_B = \frac{eh}{4\pi m} = 9.27\times10^{-24}\,\mathrm{A\,m^2}$ is called Bohr magneton which is used to measure atomic magnetic moments.

3.9 AMPÈRE'S CIRCUITAL LAW

Ampère's circuital law is used to calculate magnetic field at a point whenever there is a symmetry in the problem. This is similar to Gauss's law in electrostatics.

3.9.1 Ampère's circuital law

Ampère's law: The line integral of magnetic field over a closed loop is μ0 times net current enclosed by the loop.

$$\oint_C \vec{B}.\vec{dl} = \mu_\circ I_{enclosed} \qquad (3.51)$$

where $I_{enclosed}$ is the net current linked by the closed loop C. Note that the line integral does not depend on the shape of the path or the position of the conductor with the magnetic field.

Note

Line integral means integral over a line or curve, symbol used is $\int_c$

Closed line integral means integral over a closed curve (or line), symbol is $\oint$ *or* $\oint_c$

3.9.2 Magnetic field due to the current carrying wire of infinite length using Ampère's law

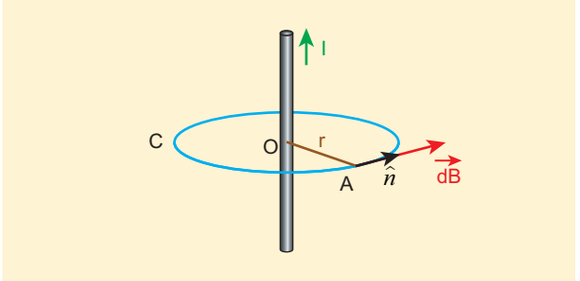


Figure 3.37 Ampèrian loop for current carrying straight wire

Consider a straight conductor of infinite length carrying current I and

12th Physics_Vol-1_Unit 3_English Medium.indd 162 23-12-2022 15:02:44